

Stochastic optimization

Exercise collection

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This collection accompanies the lecture slides. Each exercise names the deck it depends on. This version holds the *statements only*; the solutions are in `exercises_solutions.pdf`.

Sources are indicated exercise by exercise; several are adapted from Birge and Louveaux, *Introduction to Stochastic Programming*, 2nd edition, Springer, 2011.

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1 Generalities

Exercise 1.1. (Deck 03.) Show that the value of the stochastic solution and the expected value of perfect information are always non-negative.

2 Two-stage problems with recourse

Exercise 2.1. (Deck 01 and 03.) Consider the farmer problem of deck 01. Is its recourse complete? Relatively complete? Simple?

Exercise 2.2. (Deck 03; Birge and Louveaux, p. 87.) Consider the second-stage program

$$Q(x, \xi) = \min_y \{y \mid \xi y = 1 - x, y \geq 0\},$$

where ξ follows a triangular distribution on $[0, 1]$, i.e. $P[\xi \leq u] = u^2$.

- (a) Is the recourse fixed? Why?
- (b) Express $K_2(\xi)$ for every $\xi \in [0, 1]$.
- (c) Express K_2 , K_2^P and Q .

Exercise 2.3. (Deck 03.) Consider the second-stage program

$$\begin{aligned} \min_y \quad & 2y_1 + y_2 \\ \text{s.t.} \quad & y_1 - y_2 \leq 2 - \xi x_1, \\ & y_2 \leq x_2, \\ & y_1, y_2 \geq 0. \end{aligned}$$

Find $K_2(\xi)$ and K_2 when (a) $\xi \sim U[0, 1]$; (b) $\xi \sim \text{Poisson}(\lambda)$, $\lambda > 0$. Which properties do you expect of K_2 ? Is the recourse simple or complete? If not, under which conditions would it be relatively complete?

Exercise 2.4. (Deck 03; Birge and Louveaux, p. 102.) Consider the second-stage program

$$\begin{aligned} \min_y \quad & 2y_1 + y_2 \\ \text{s.t.} \quad & y_1 + y_2 \geq 1 - x_1, \\ & y_1 \geq \xi - x_1 - x_2, \\ & y_1, y_2 \geq 0. \end{aligned}$$

Show that this program has a complete recourse when ξ has a finite expectation.

Exercise 2.5. (Deck 03.) A company must order a quantity x of a product to meet a demand d that is unknown at the time of ordering. The unit ordering cost is $c > 0$. If $d > x$, a penalty $b \geq 0$ is incurred per unit of unmet demand; if $d < x$, a holding cost h is incurred per surplus unit.

- (a) Formulate the problem as a two-stage stochastic program.
- (b) Is the recourse simple, complete, relatively complete?
- (c) What can be expected of the first-stage solution when $b < c$?

Exercise 2.6. (Deck 03.) A good X must be produced before its demand is known. Each unit produced costs \$2. Demand is discrete, taking the values D_s with probabilities p_s , $s = 1, \dots, S$. Demand can also be met by buying from an external supplier at \$3 per unit, once it has been observed. How much should be produced now? Formulate the problem as a two-stage stochastic program minimizing the expected production cost.

Exercise 2.7. (Deck 03; Higle, Tutorials in Operations Research, INFORMS, 2005.) The Dakota Furniture Company makes desks, tables and chairs. Each item consumes lumber and two kinds of skilled labour, finishing and carpentry:

Resource	Cost	Desk	Table	Chair
Lumber (board ft.)	\$2	8	6	1
Finishing (hours)	\$4	4	2	1.5
Carpentry (hours)	\$5.2	2	1.5	0.5
Selling price		\$60	\$40	\$10

Resources must be acquired before the demand is known. Three equally structured demand scenarios are considered, with probabilities 0.3, 0.4 and 0.3:

Scenario	Desks	Tables	Chairs
Low ($p = 0.3$)	50	20	200
Most likely ($p = 0.4$)	150	110	225
High ($p = 0.3$)	250	250	500

Formulate the problem as a two-stage stochastic program maximizing the expected profit. Only the formulation is asked.

Exercise 2.8. (Deck 03; Birge and Louveaux, p. 142.) Consider

$$\begin{aligned} \min_x \quad & x_1 + 4x_2 + \mathbb{E}_{\xi}[Q(x, \xi)] \\ \text{s.t.} \quad & x_1 + x_2 = 1, \quad x \geq 0, \end{aligned}$$

with

$$\begin{aligned} Q(x, \xi) = \min_y \quad & y_1 + 10y_2 + 10y_3 \\ \text{s.t.} \quad & y_1 + y_2 - y_3 = \xi + x_1 - 2x_2, \\ & y_1 \leq 2, \quad y \geq 0, \end{aligned}$$

and $\xi \sim U[1, 3]$.

(a) Is the recourse fixed, simple, complete, relatively complete? Answer true or false for each, with a brief justification.

(b) For given x and ξ , show, by discussing the value of $\xi + x_1 - 2x_2$, that

$$y^*(x, \xi) = \begin{cases} y_1 = \xi + x_1 - 2x_2, y_2 = y_3 = 0 & \text{if } 0 \leq \xi + x_1 - 2x_2 \leq 2, \\ y_1 = 2, y_2 = \xi + x_1 - 2x_2 - 2, y_3 = 0 & \text{if } \xi + x_1 - 2x_2 > 2, \\ y_3 = 2x_2 - \xi - x_1, y_1 = y_2 = 0 & \text{if } \xi + x_1 - 2x_2 < 0. \end{cases}$$

(c) Which expected property of $Q(x, \xi)$ does $y^*(x, \xi)$ confirm?

(d) Is $Q(x) = \mathbb{E}_{\xi}[Q(x, \xi)]$ differentiable? Why?

(e) Express K_1 , $K_2(\xi)$, K_2^P and K_2 .

(f) Give the deterministic equivalent (extensive form) of the discretized problem where ξ takes the values 1, 2, 3.

3 Decomposition methods

Exercise 3.1. (Deck 04.) Consider

$$\begin{aligned} \min & 100x_1 + 150x_2 + \mathbb{E}_{\xi}[q_1y_1 + q_2y_2] \\ \text{s.t. } & x_1 + x_2 \leq 120, \\ & 6y_1 + 10y_2 \leq 60x_1, \quad 8y_1 + 5y_2 \leq 80x_2, \\ & y_1 \leq d_1, \quad y_2 \leq d_2, \\ & x_1 \geq 40, \quad x_2 \geq 20, \quad y_1, y_2 \geq 0, \end{aligned}$$

under two scenarios $\xi = (d_1, d_2, q_1, q_2)$: $\xi_1 = (500, 100, -24, -28)$ with probability $p_1 = 0.4$, and $\xi_2 = (300, 300, -28, -32)$ with probability $p_2 = 0.6$.

Run the first iteration of the single-cut L-shaped method, showing that the second-stage solutions are respectively $y^* = (137.5, 100)$ and $y^* = (80, 192)$. To obtain the dual solutions, use complementary slackness: a dual variable can be non-zero only if the associated primal constraint is active.

Exercise 3.2. (Deck 04.) In the regularized decomposition method of Ruszczyński, what is the difference between an exact serious step and an approximate serious step? Why must the two be distinguished?

4 Multistage problems

Exercise 4.1. (Decks 06 and 08; Heitsch, Römisch and Strugarek, Stability of multistage stochastic programs, *SIAM Journal on Optimization* 17(2):511–525, 2006.) Consider the stylized multistage program below, seeking an optimal purchasing strategy under uncertain costs ξ_t over the periods $t = 0, \dots, T$. The amounts purchased are x_t , and a prescribed amount a must be reached at the horizon T :

$$\begin{aligned} \min_x & \mathbb{E} \left[\sum_{t=0}^T \xi_t x_t \right], \\ \text{s.t. } & s_t - s_{t-1} = x_t, \quad t = 0, \dots, T, \\ & s_{-1} = 0, \quad s_T = a, \quad x_t \geq 0, \quad s_t \geq 0, \end{aligned}$$

where s_t is a state variable holding the amount accumulated at time t .

Take three stages ($t = 0, 1, 2$), a binary tree, and $\xi_0 = 10$; the cost goes up by 1 with probability 0.6 and down by 1 with probability 0.4.

(a) Draw the scenario tree.

(b) Write the problem (i) in extensive form, (ii) using a scenario decomposition.

(c) In which of the two formulations must non-anticipativity constraints be written explicitly?

5 Random number generation

Exercise 5.1. (Deck 02.) A linear congruential generator produces $x_i = (ax_{i-1} + c) \bmod m$.

- (a) For $m = 16$ and $c = 1$, use the period theorem to find every a giving a full period.
- (b) The “standard minimal” generator of Park and Miller is $x_i = 16807 x_{i-1} \bmod (2^{31} - 1)$. Why can its period not be m ? What is the largest it could be, and why is that value actually reached here?

Exercise 5.2. (Deck 02.) Consider the linear feedback shift register over $\mathbb{F}_2$ with $k = 4$ and $x_n = x_{n-1} \oplus x_{n-4}$, seeded at 0001.

- (a) Why is the all-zero state excluded, and why can the period not exceed $2^k - 1$?
- (b) A Tausworthe generator reads the output stream w bits at a time, spaced ν apart. The slides take $w = 3$, $\nu = 3$ and observe only five distinct values instead of fifteen. Explain, and find every $\nu \in \{1, \dots, 14\}$ giving the maximal output period.

Exercise 5.3. (Deck 02.) Let X take the values 1, 2, 3 with probabilities 0.2, 0.5, 0.3.

- (a) Write F_X and its generalized inverse, and check that $F_X^{-1}(U)$, $U \sim U[0, 1]$, has the right distribution.
- (b) Why can the ordinary inverse not be used here?
- (c) Inversion is called the “gold standard” in the slides. Which property makes common random numbers and antithetic variates reliable with it?

Exercise 5.4. (Deck 02.) In acceptance–rejection one takes $t \geq f$ with $\int t = a$, so that $r = t/a$ is a density, then repeats: draw $Y \sim r$ and $U \sim U[0, 1]$ until $U \leq f(Y)/t(Y)$, and return Y .

- (a) Compute the probability that one iteration is accepted.
- (b) Deduce the expected number of iterations, and explain why the slides ask for a close to 1.
- (c) Why does this method break common random numbers, even when the accepted value is correct?

Exercise 5.5. (Deck 02.) Let C be a bivariate copula, so that $H(x, y) = C(F_X(x), F_Y(y))$ is the joint cdf of (X, Y) .

- (a) Show that the independence copula $C(u, v) = uv$ satisfies the Fréchet–Hoeffding bounds.
- (b) The upper bound is attained by $C(u, v) = \min(u, v)$. Which dependence between X and Y does it describe?
- (c) A tempting proof of the upper bound argues $P[A \cap B] \leq P[A] P[B] \leq \min(P[A], P[B])$. Give a counterexample to the first inequality, and state the correct argument.

6 Monte Carlo approximation

Exercise 6.1. (*Decks 03 and 09.*) Show that

$$\mathbb{E}_{\boldsymbol{\xi}} \left[\min_{x \in \mathcal{X}} z(x, \boldsymbol{\xi}) \right] \leq \min_{x \in \mathcal{X}} \mathbb{E}_{\boldsymbol{\xi}} [z(x, \boldsymbol{\xi})]$$

for a deterministic feasible set $\mathcal{X}$. How is this result to be interpreted for stochastic programming? Extend it to a Monte Carlo sample average.